

LEON DO

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Education

California State University, Long Beach

Expected May 2027

B.S. in Computer Science, Minor in Mathematics

Long Beach, CA

- Relevant Coursework: Systems Programming, Computer Architecture, Networks & Distributed Computing, Data Structures & Algorithms, Optimization
- CSULB Programming Team; competed in ICPC SoCal 2025

Experience

Zwei Labs

Oct 2025 - Jan 2026

Software Engineering Intern

Remote

- Developed a real-time Python trading backend for market data, live order books, strategy execution, and pre-trade risk checks.
- Scaled to 40+ concurrent markets using independent async workers and WebSocket streams, isolating slow feeds from unrelated markets.
- Added bounded queues, backpressure, deterministic event ordering, and exact decimal pricing to preserve correctness under load.

Beach Investment Group

Jun 2024 - May 2025

Quantitative Developer Intern

Long Beach, CA

- Created a scalable Python/Polars data pipeline used by 25+ analysts for ingestion, preprocessing, model training, and prediction delivery.
- Accelerated NumPy with CuPy / CUDA, cutting experiment times from hours to minutes and increasing research throughput.
- Implemented reusable factor-model and time-series tools with scikit-learn; forecasts supported funding proposals that secured \$115K.

Projects

Monte Carlo Benchmarking Engine (C++17, AVX2)

- Engineered a C++17 performance engine with AVX2 SIMD, multithreading, thread-local state, and preallocated memory.
- Increased throughput 11x, processing 100M trials in 106 ms versus 1.19 seconds sequentially.
- Profiled CPU performance with Linux perf, tracking IPC, cache misses, branch misses, and cycles per trial.

MaguroPlace - GPU-Accelerated Placement Optimizer (Python, PyTorch, CUDA)

- Created a GPU-accelerated optimization engine in Python/PyTorch that produced valid layouts across all 17 benchmarks.
- Improved evaluation throughput 18x, from 11 minutes to 38 seconds, by profiling bottlenecks and caching reusable benchmark data.
- Reduced the mean challenge score 47% (2.09 to 1.10), beating the published 1.46 baseline.

Tori - Concurrent Monitor (Go)

- Designed a concurrent Go service with isolated workers, retry/backoff, and clean context cancellation.
- Kept memory usage predictable under load with backpressure, bounded queues, and rate limiting.
- Added observability with Go pprof and runtime metrics for CPU, memory, contention, garbage collection, and worker health.

DeepMew - Resumable Cloud Jobs (Python, PyTorch, Modal)

- Ran distributed PyTorch workloads on Modal cloud workers with checkpointing, restart recovery, and cost controls for fault tolerance.
- Improved reliability by fixing a checkpoint bug that erased progress after worker restarts; verified the fix across more than 400K games.
- Scaled a rate-limited data pipeline from 1,543 replays to 180K training rows while staying within API limits.

Skills

Languages: C++, Python, Go

Systems: Linux, Systems Programming, Distributed Systems, Concurrency, Scalability, Reliability, Async I/O

Infrastructure: Docker, Kubernetes, AWS, Redis, Observability

Performance / Data: Multithreading, Performance Profiling, SIMD/AVX2, CUDA, PyTorch, CuPy, Polars, ClickHouse, Parquet